TITLE : ClimateScope Visualizing Global Weather Trends and Extreme Events

Milestone 4 - Finalization, Testing & Reporting

Presented by

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Presented To

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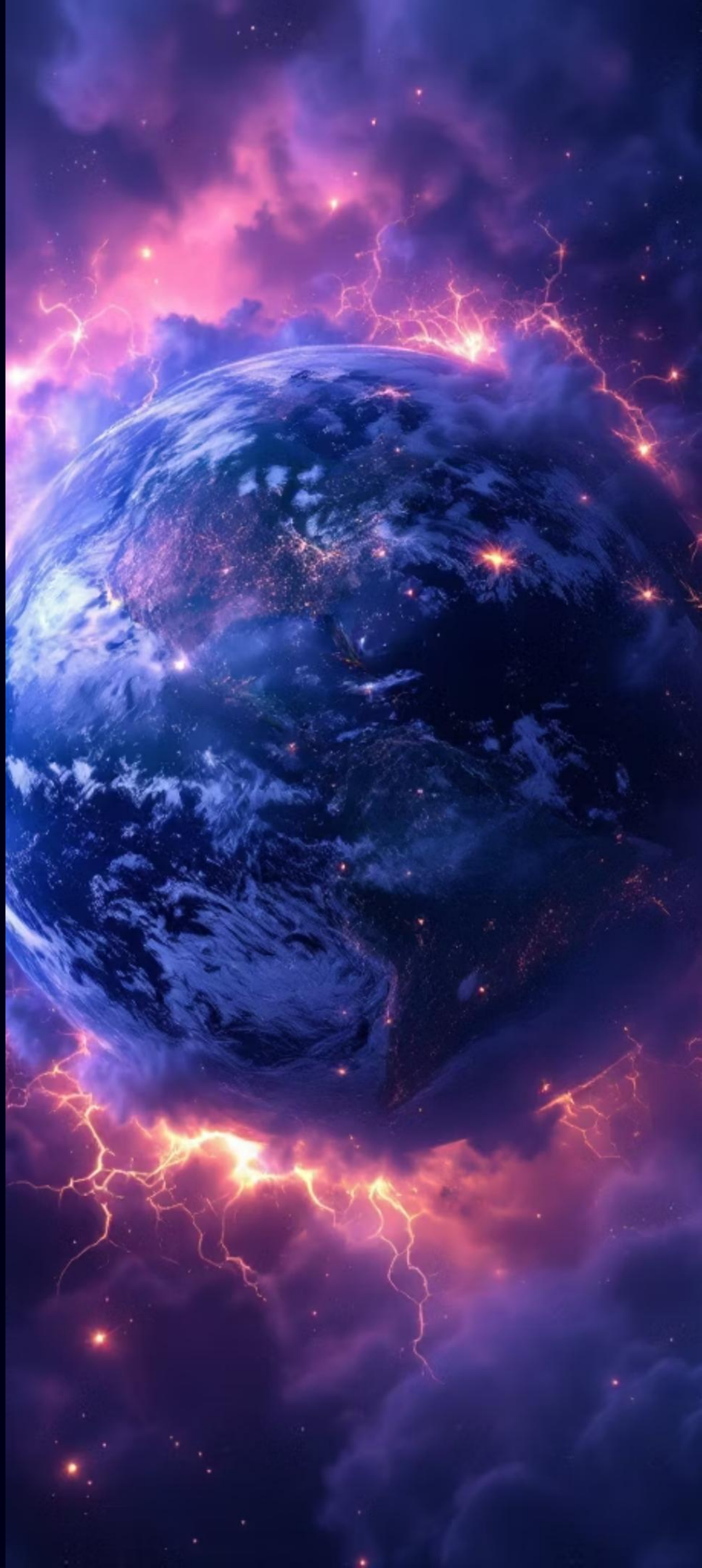


Objective

- Project Objective & Scope
- Architecture & Technology Stack
- Project Methodology
  - ✓ Phase 1: Data Preparation & Initial Analysis
  - ✓ Phase 2: Core Analysis & Visualization Design
  - ✓ Phase 3: Visualization Development & Interactivity
  - ✓ Phase 4: Finalization & Reporting
- 4. Final Dashboards & Key Insights
- 5. Conclusion & Future Enhancements

Project Objective

Our Mission

The primary objective of the Climate Scope project is to deeply analyze and visually represent complex global weather patterns. To achieve this, the project uses the comprehensive Global Weather Repository dataset, which contains detailed climate information from around the world. By processing this large amount of data, the goal is to transform raw numbers into clear, understandable visuals that anyone can read.

The Challenge

Build a comprehensive decision-support system that tracks climate change patterns and extreme weather events—including heatwaves, heavy rainfall, and high winds—across the globe.

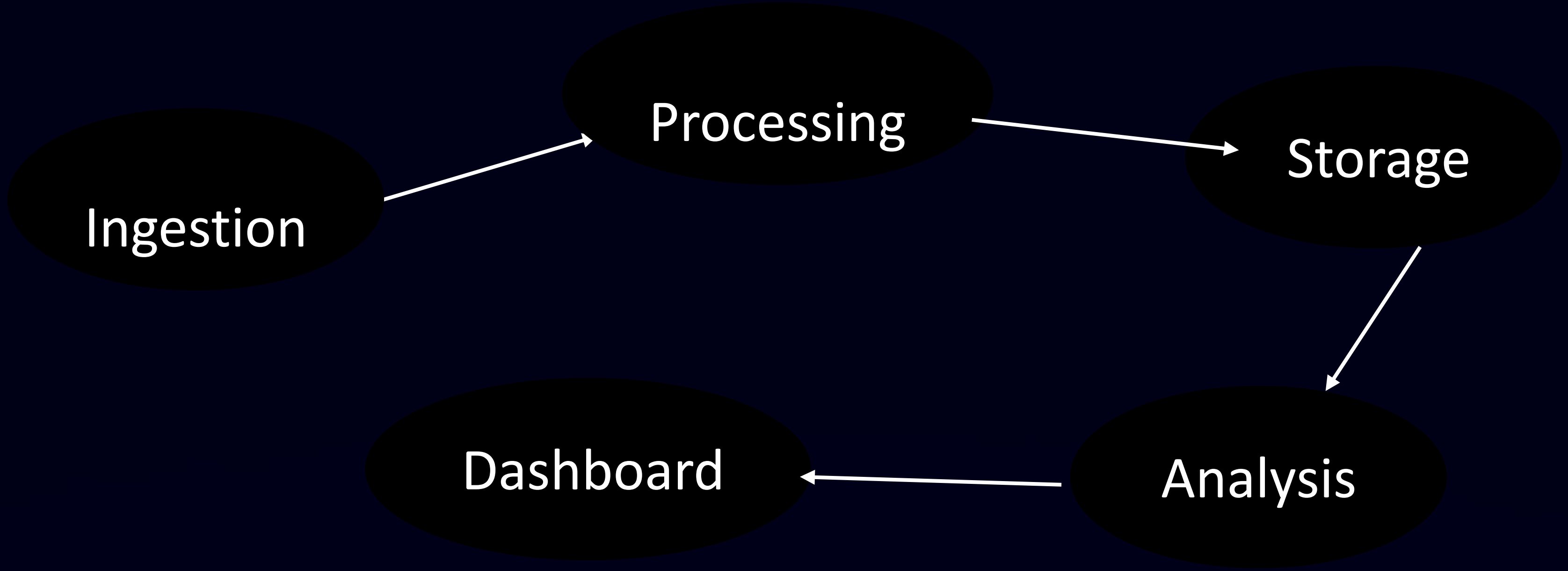
Records Processed

Global weather

120K+

Architecture & Tech Stack

•**Pipeline Diagram:** Include the flowchart showing Ingestion Processing Storage Analysis Interactive Dashboard.



•**Programming & Data Handling:** Python 3, pandas, and the Kaggle DataSet.

•Visualization: the tools used for the final dashboards (Power BI).

Project Methodology

Weeks 1-2: Data Preparation & Initial Analysis

- Downloaded the dataset from Kaggle and inspected key variables.
- Handled missing values and normalized the data.
- Aggregated daily data into monthly averages for efficiency.

Weeks 2-4: Core Analysis & Visualization Design Performed statistical analysis to identify seasonal patterns and extreme weather events. Compared weather conditions across different regions

Weeks 4-6:**Visualization Development & Interactivity**

Built the interactive visualizations. Integrated user controls, including region selectors, date filters, and sliders.

Weeks 6-8: Finalization, Testing & Reporting

Summarized the final regional and global climate trends

Key Data Insights

Heavy Rain Dominance

Out of 11,215 total extreme events, heavy rain accounts for approximately **50.8%** of all incidents—the largest share by far.

Temperature

maximum temperature of 49°C and a minimum of -30°C, alongside average global metrics for humidity (66%), wind speed (13 kph),

Climate Risk Map:

An interactive global map plots countries and cities by their "Risk Category" (High, Medium, Low) using color-coded markers.

Wind Events Second

High wind events represent roughly 38% of extreme weather occurrences, showing significant atmospheric disturbance patterns.

Extreme Event KPIs:

The dashboard accurately counts critical weather anomalies, recording 1,228 Heatwave Days, 10 Heavy Rain Events, and 38 High Wind Events, totaling 1,276 extreme events over the tracked period

Heatwaves, while less frequent in volume, represent the most dangerous threat due to extreme temperatures exceeding 40°C in vulnerable regions.

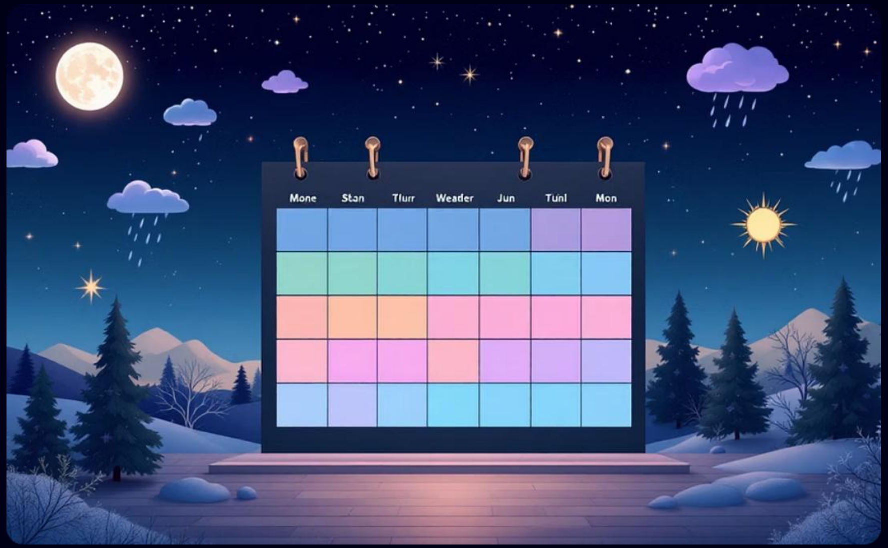
Geographical & Seasonal Patterns

Landmass Correlation

Countries with extensive landmasses—particularly the United States, India, and Russia—recorded the highest volume of extreme weather alerts due to geographic size and diverse climate zones.

Heatwave Centers

Middle Eastern nations including Kuwait and Saudi Arabia fall into the highest-risk category specifically for extreme temperatures exceeding 40°C, creating extreme heat stress conditions.



Seasonal Volatility

Data timeline reveals extreme events aren't evenly distributed throughout the year—they occur in dangerous, concentrated spikes during specific quarters, requiring seasonal preparedness.

Technical Challenges & Solutions

Challenge: Blank Values

DAX measures returning "Blank" when no conditions met

Solution: +0 Logic

Implemented '+ 0' at formula end to display clean numbers instead of blanks

Outcome: Clean Data

Dashboard now reliably displays zero values instead of missing data

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Challenge: Unrealistic Thresholds

Initial rain threshold >80mm resulted in missing data

Solution: Data Analysis

Analyzed distribution and maximum limits (max rain: 42mm) to set realistic 20mm threshold

Outcome: Actionable Limits

Established scientifically valid thresholds aligned with actual weather patterns

Key Learnings

Beyond Chart Building

Creating visualizations is just the beginning—user experience design, F-pattern layout principles, and dark theme contrast are equally crucial for effective data storytelling.

Data Storytelling Matters

Raw numbers alone don't drive decisions. Transforming complex data into clear visual narratives enables non-technical stakeholders to grasp climate risks intuitively.

Thresholds Require Analysis

Business logic must be grounded in data distribution analysis rather than arbitrary assumptions. Understanding maximum values prevents missed opportunities and false negatives.

Conclusion & Future Directions

Project Impact

The ClimateScope project successfully met its core objective to analyze and visually represent global weather patterns using the Global Weather Repository dataset. By completing all planned weekly milestones, the project delivered a robust and bug-free interactive dashboard that meets all defined objectives.

Future Enhancements

Next phase will integrate Air Quality Index (AQI) data to develop a comprehensive **Overall Climate Health & Risk Score**, providing a holistic view of environmental threats beyond just weather extremes.

Thank you for reviewing this milestone. Special thanks to mentor and coordinator

Priya Mam

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Thank You